Smart Attendance System (SAS)

A project report submitted in partial fulfillment of the requirements for the award of the degree of

B.Tech. in Computer Science

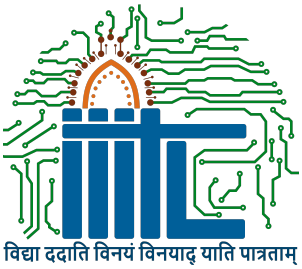
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Declaration of Authorship

We, **Pranilash, Nithin, Radesh, Ranadeep, and Tanishq**, declare that the work presented in **"Smart Attendance System (SAS)"** is our own. We confirm that this project was completed as part of the B.Tech. program at the Indian Institute of Information Technology, Lucknow. Wherever we have consulted the published work of others, the sources have been cited appropriately.

Signed:

Date:

Certificate

This is to certify that the project entitled **"Smart Attendance System (SAS)"** submitted by **Pranilash, Nithin, Radesh, Ranadeep, and Tanishq** for the award of the B.Tech. degree at the Indian Institute of Information Technology, Lucknow is based on their own work carried out under the supervision of **Dr. Rahul Kumar Verma**. This work has not been submitted elsewhere for any degree or diploma.



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Acknowledgment

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Abstract

Smart Attendance System (SAS) is a secure web-based attendance platform developed to reduce proxy attendance and automate attendance recording in educational institutions. The system combines rotating QR codes, face verification, and liveness checks to ensure that attendance is marked only for authenticated and physically present students.

The application uses a role-based design with separate dashboards for administrators, faculty members, and students. Faculty members can create classes, start sessions, and monitor live attendance. Students first register their face using the webcam and then mark attendance by scanning the session QR code and completing face verification. Blink and head-movement checks improve resistance against spoofing attempts.

The project is implemented using the MERN stack. React, Vite, Tailwind CSS, and face-api.js are used on the frontend, while Node.js, Express.js, MongoDB, JWT authentication, and validation middleware are used on the backend. The system also provides attendance history, reports, and CSV export for better record management. Early testing shows that the workflow is practical for classroom deployment and can be extended with BLE-based proximity verification in future work.

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Chapter 1

Introduction

Attendance management is an important part of academic administration, but it is often affected by manual errors, proxy attendance, and delays in record keeping. In large classrooms, it is difficult for faculty members to verify the physical presence of every student with complete certainty. Traditional methods such as paper registers and swipe-card systems are easy to use, but they do not provide strong identity verification and can be misused.

The Smart Attendance System (SAS) is designed to address these issues by combining rotating QR-code based session initiation with face verification and liveness checks. The system ensures that attendance is marked only when a student is authenticated during an active faculty-controlled session. Since the application is web-based, it can be accessed through standard browsers without requiring specialised attendance hardware.

1.1 Problem Statement

The objective of the project is to design and implement a secure attendance system that reduces proxy attendance while remaining simple to use for faculty and students. The proposed solution should support real-time verification, maintain centralized attendance records, and work reliably in a classroom environment.

1.2 Need for the System

The proposed system is needed because conventional attendance methods do not fully solve the problem of attendance fraud. Manual registers are slow, QR-only systems can be shared, and face-only systems can be spoofed using images or videos. A secure attendance platform is therefore required to combine identity verification, session control, and real-time attendance tracking in one workflow.

1.3 Objectives

The main objectives of the project are:

1. To provide role-based access for administrators, faculty members, and students.
2. To generate rotating QR codes for active attendance sessions.
3. To verify student identity using real-time face recognition.
4. To include liveness detection for reducing spoofing attacks.
5. To store attendance history and generate reports for analysis.
6. To support a scalable and web-based attendance management workflow.

1.4 Scope of the Project

The proposed system supports class creation, session management, face registration, attendance marking, and reporting. It is designed as a web application so that students can use a laptop or mobile camera and faculty can monitor attendance live from any supported browser.

1.5 Applications

The Smart Attendance System can be used in:

- colleges and universities,
- coaching institutes,
- training centres,
- seminar and workshop sessions, and
- any classroom-based attendance environment that requires secure verification.

1.6 Technology Overview

The application is built using the MERN stack. React and Vite are used for the frontend, Express and Node.js handle the backend APIs, and MongoDB stores users, classes, sessions, embeddings, and attendance records. JWT-based authentication, input validation, and security middleware help protect the system from unauthorized access and common web attacks.

1.7 Chapter Summary

This chapter introduced the attendance problem, the motivation for the project, and the main goals of the Smart Attendance System. The next chapter reviews related work and the technologies used in similar attendance solutions.

Chapter 2

Literature Review

This chapter reviews existing attendance management approaches and the technologies that influenced the proposed Smart Attendance System. The survey focuses on QR-based systems, biometric verification, face recognition, liveness detection, and secure web-based implementation models [1, 2, 3, 4, 5, 6, 7, 8].

2.1 QR Code-Based Attendance Systems

QR-based attendance systems are popular because they are simple, inexpensive, and easy to deploy. A faculty member can generate a QR code quickly, and students can scan it using a phone or browser. However, standard QR systems are vulnerable to screenshot sharing and remote proxy attendance. For that reason, recent work recommends short-lived or dynamically changing QR codes [2, 4].

2.2 Face Recognition and Biometric Verification

Face recognition has become a practical biometric method for attendance systems because it is contactless and works with ordinary webcams. Deep-learning-based facial embeddings can identify users accurately when the captured image is clear and properly aligned [1, 3]. The main limitation is that face-only systems may still be fooled by photographs, video replays, or screen-based attacks.

2.3 Liveness Detection Techniques

Liveness detection is used to confirm that the captured face belongs to a real person rather than a static image. Common techniques include blink detection, head-movement analysis, and motion-based checks. These methods improve security and are especially useful in educational settings where attendance fraud is common [4].

2.4 Secure Web-Based Attendance Platforms

Modern attendance systems increasingly use web technologies because they are easier to maintain and can support centralized dashboards, analytics, and role-based access. React, Node.js, Express, MongoDB, and JWT are widely used for scalable web applications [6, 7, 8]. Security practices such as validation, rate limiting, and token-based authentication are important in these systems.

2.5 Integrated Attendance Solutions

Recent research shows that combining QR authentication with facial verification produces a stronger attendance workflow than using either method alone. Hybrid systems improve usability while reducing proxy attendance and increasing confidence in the attendance record [1, 2, 3]. This approach directly influenced the design of the proposed system.

2.6 Comparative Analysis of Related Solutions

Table 2.1 summarizes the strengths of common attendance approaches and shows how the proposed system combines their advantages.

Criteria	**Manual**	**RFID**	**QR Only**	**Face Only**	**SAS**
Fast attendance marking	No	Yes	Yes	Yes	Yes
Proxy attendance prevention	No	No	No	Partial	Yes
Contactless operation	No	Yes	Yes	Yes	Yes
Low hardware requirement	Yes	No	Yes	Yes	Yes
Real-time verification	No	Partial	Partial	Partial	Yes
Liveness / spoof detection	No	No	No	Partial	Yes

Table 2.1: Comparison of attendance approaches

2.7 Research Gap

The review shows that many systems solve only one part of the attendance problem. Manual systems are inefficient, RFID systems can be shared, QR-only systems can be copied, and face-only systems can be spoofed. The gap addressed by this project is

the need for a single web-based solution that combines secure QR verification, face matching, and liveness detection in a practical workflow.

2.8 Chapter Summary

The literature indicates that a hybrid attendance system is more reliable than a single-method solution. This finding supports the design of the Smart Attendance System presented in the next chapter.

Chapter 3

Methodology and System Design

This chapter explains the design and working of the Smart Attendance System. The solution follows a modular architecture so that authentication, attendance verification, and reporting can be handled independently while still working together as one system.

3.1 System Architecture

The system uses a three-tier design.

- **Presentation Layer:** React-based dashboards for administrators, faculty members, and students.
- **Application Layer:** Express.js REST APIs for authentication, QR validation, face verification, attendance processing, and reporting.
- **Data Layer:** MongoDB collections for users, classes, sessions, embeddings, and attendance records.

Figure 3.1 illustrates the overall flow.

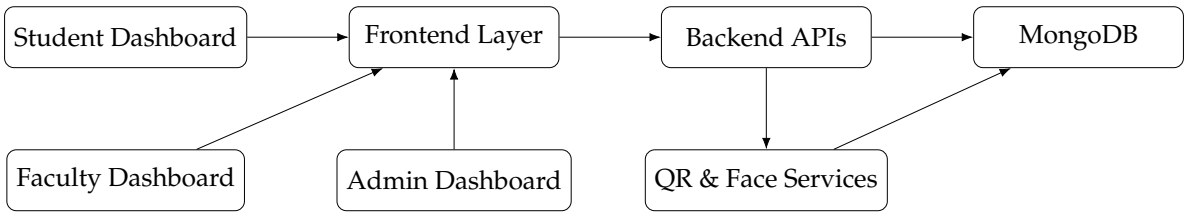


Figure 3.1: System architecture of the Smart Attendance System

3.2 Workflow

The attendance workflow is designed to verify both session validity and student identity.

1. Faculty logs in and creates or selects a class.
2. Faculty starts an attendance session.
3. The system generates a rotating, signed QR code.
4. Student scans the QR code to request verification.
5. The student completes face verification through the webcam.
6. If the face matches and liveness checks pass, attendance is recorded.
7. Faculty can view the live count and export attendance reports.

3.3 Core Modules

3.3.1 Authentication and Role Management

Authentication is implemented using JSON Web Tokens (JWT). Access tokens and refresh tokens are used to maintain secure user sessions. Role-based access control mechanisms ensure that users can access only the functionalities assigned to their respective roles.

The system supports three primary user roles:

- Administrator
- Faculty Member
- Student

3.3.2 QR Attendance Module

The QR attendance module generates dynamic and time-sensitive QR codes during active attendance sessions. Each QR code contains encrypted session information signed using HMAC hashing techniques. The short refresh interval prevents unauthorized sharing and reuse.

3.3.3 Face Registration and Verification

Students are required to register their facial data before participating in attendance sessions. During registration, facial embeddings are generated using the `face-api.js` library and stored securely in the database. During attendance verification, the live facial image captured through the webcam is compared with the stored embedding using cosine similarity calculations.

3.3.4 Liveness Detection

Liveness detection mechanisms are incorporated to prevent spoofing attacks using photographs, videos, or digital screens. The system performs basic liveness checks such as eye blink detection, head movement verification, and real-time motion analysis.

3.3.5 Administration and Reporting Module

The administration module allows administrators to manage departments, faculty members, students, and system-level configurations. The faculty module provides class creation, attendance session control, live attendance monitoring, report generation, and CSV export functionality.

3.4 Entity Relationship Design

The database is organized around the academic entities used in the system. Administrators manage users, faculty members create and run sessions, and students participate in attendance marking through QR and face verification. Each department can contain multiple classes, each class can have multiple attendance sessions, and each attendance session can generate multiple attendance records. A student may have one stored face embedding for verification, while faculty members manage classes and attendance sessions.

Figure 3.2 shows the ER style design used for the project.

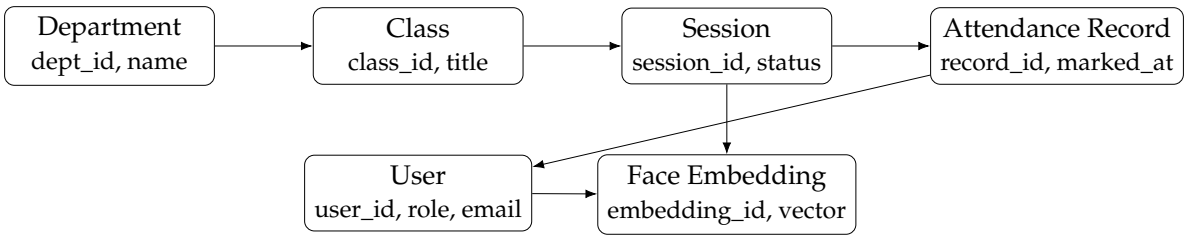


Figure 3.2: ER diagram of the Smart Attendance System

3.5 Database Design

The MongoDB database consists of users, departments, classes, attendance sessions, attendance records, and face embeddings. The schema is designed to support efficient retrieval of live session data and historical attendance reports.

3.6 Security Considerations

The Smart Attendance System incorporates multiple security layers to protect user data and prevent unauthorized access.

- Helmet middleware for secure HTTP headers.
- CORS restrictions to block untrusted origins.
- Rate limiting to reduce brute-force and spam requests.
- bcrypt hashing for password protection.
- Zod validation for request sanitization.
- HMAC-signed QR payloads to protect integrity.
- Secure JWT cookies for session handling.

3.7 Algorithm for Attendance Verification

Algorithm 1 Attendance verification process

```
Scan QR code from the active session
Validate the QR signature and session timestamp
Capture webcam frame for face verification
Generate facial embedding for the captured image
Compare the embedding with the stored student embedding
Perform liveness checks such as blink and movement detection
if all checks pass then
    Mark attendance and update records
else
    Reject the request and show an error message
end if
```

3.8 Chapter Summary

The proposed design combines QR-based verification, biometric authentication, and liveness detection within a secure web architecture. The modular design improves scalability, maintainability, and practical usability for academic attendance management.

Chapter 4

Implementation, Deployment, and Results

This chapter discusses the implementation details, deployment process, testing procedures, and observed results of the Smart Attendance System. The system was developed as a full-stack web application using modern technologies and security mechanisms to ensure efficient and secure attendance management.

4.1 Implementation Stack

The Smart Attendance System is implemented using the MERN stack along with additional libraries for face recognition, QR code processing, and security.

4.1.1 Frontend Technologies

- **React 18** – Component-based user interface.
- **Vite** – Fast development and production builds.
- **Tailwind CSS** – Responsive interface styling.
- **Zustand** – Global state management.
- **Axios** – API communication.
- **react-router-dom** – Route navigation.
- **face-api.js** – Face detection and facial embeddings.
- **html5-qrcode** – QR scanning.

4.1.2 Backend Technologies

- **Node.js** – Server-side runtime.
- **Express.js** – REST API routing.
- **MongoDB with Mongoose** – Database and schema layer.
- **JWT Authentication** – Secure login sessions.
- **Security Middleware** – Helmet, CORS, and rate limiting.
- **Zod Validation** – Request validation and sanitization.

4.2 Database Design

MongoDB is used as the primary database. The major collections are:

- Users – administrator, faculty, and student profiles.
- Departments – department information.
- Classes – class and subject details.
- Attendance Sessions – active and completed sessions.
- Attendance Records – attendance history.
- Face Embeddings – biometric vectors for verification.

4.3 API Design

The backend exposes RESTful endpoints for authentication, class and session management, face registration, QR verification, attendance marking, and reporting. The API structure is separated into authentication, admin, faculty, and student routes so that responsibilities remain clear and maintainable.

4.4 Deployment Strategy

The project can run in local development mode or be deployed to a production server. Environment variables configure the database connection, JWT secrets, CORS settings, and optional services such as Redis or Cloudinary. The backend can run directly with Node.js or inside Docker containers, while the frontend can be served through a static hosting setup.

4.5 Testing and Validation

The system was validated through the following checks:

- login and role authorization,
- QR generation and expiry,
- face registration and matching,
- liveness detection,
- attendance marking flow,
- report generation and CSV export.

Table 4.1 summarizes the testing outcomes.

Test Case	**Expected Result**	**Observed Result**
Invalid QR scan	Request rejected	Passed
Valid QR with wrong face	Attendance rejected	Passed
Valid QR with registered face	Attendance marked	Passed
Expired session QR	Request rejected	Passed
CSV export from faculty panel	File downloaded correctly	Passed

Table 4.1: Functional testing summary

4.6 Results and Discussion

The implemented solution reduced the time needed to mark attendance and improved attendance reliability. The rotating QR code limited code reuse, face verification improved identity validation, and liveness checks reduced spoofing attempts. Faculty members could monitor live attendance, and students could view attendance history from their own dashboards.

4.7 Advantages of the Proposed System

The main advantages of the system are:

- reduced proxy attendance,
- contactless verification,
- real-time monitoring,

- centralized attendance records,
- role-based access control,
- and easy deployment in academic environments.

4.8 Chapter Summary

The implementation results show that the proposed design is practical and effective for attendance management in educational institutions. The final chapter summarizes the contribution of the project and outlines future improvements.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

The Smart Attendance System demonstrates that attendance management can be made more secure and efficient by combining rotating QR codes, face verification, and liveness checks in one web-based platform. The role-based design supports administrators, faculty members, and students without making the workflow complicated.

The project meets its main objectives by reducing manual effort, improving attendance accuracy, and limiting proxy attendance. Faculty members can control sessions and generate reports, while students can mark attendance quickly through a browser-based interface.

5.2 Limitations

The system depends on camera quality, lighting conditions, and browser support. Face verification may also require threshold tuning for different environments, and internet connectivity is needed when remote assets or services are used.

5.3 Future Work

Possible improvements include mobile application support, offline synchronization, stronger anti-spoofing models, Bluetooth Low Energy (BLE) based proximity verification, automatic notifications, attendance trend analytics, and integration with institutional ERP systems.

5.3.1 BLE-Based Proximity Verification

A possible future enhancement to the Smart Attendance System is the integration of Bluetooth Low Energy (BLE) for proximity verification. In this approach, the faculty

member's device can act as a BLE beacon during an active attendance session. Students will be allowed to proceed with attendance only when their device detects the authorized beacon within a valid proximity range.

This feature can reduce location-based proxy attendance and strengthen the existing QR and face verification flow. The BLE layer will ensure that attendance is marked only when the student is physically present near the faculty member's device. This enhancement is especially useful in large classrooms, laboratories, and seminar halls where physical proximity needs to be confirmed more strictly.

5.4 Final Remark

The project provides a practical foundation for secure digital attendance systems and can be extended further for larger academic deployments.

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